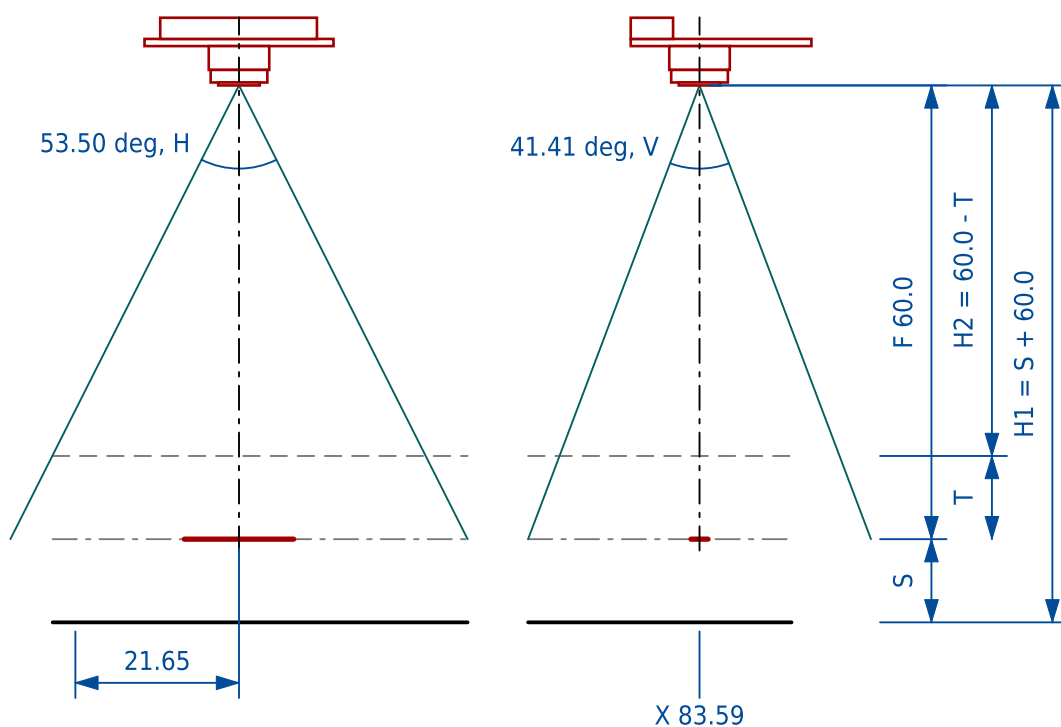
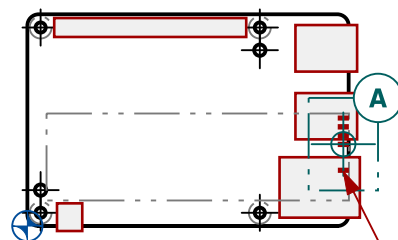


END ELEVATION

FRONT ELEVATION



PLAN, 1:2



NOTES

1. First angle; the end elevation is seen from the right. One camera: a Raspberry Pi Camera Module v1.3, stock 65 deg lens, looking straight down, reading the whole sensor.
2. Heights are to the lens FACE. The entrance pupil the optics are worked to is ASSUMED behind it by at most the lens's 5.20 mm, so the picture is up to 9.2% larger, never smaller, and the LEDs no nearer than the close limit.

LEDs A1-A4 and PWR

NOTES (continued)

3. F 60.0 is the higher of what frames frame A focused there, $24.3 + f 3.60 = 27.9$ (DERIVED, thin lens), and the lens's close limit unscrewed, 60 mm: "About the closest you can get this way is about 6 cm." (a Raspberry Pi forum user, 2013; REPORTED).
4. $H1 = S + 60.0$, where S is the mounting plate's face, the base of the standoffs under the Pi 5, to the card's top face: MEASURE, not published.
5. $H2 = 60.0 - T$, the clearance, where T is the card's top face to the assembly's highest point, the heatsink's blower or a part of the HAT: MEASURE, not published.
6. REFOCUS THE LENS TO F. As sold it is set far -- "Approx 1 m to infinity" (Raspberry Pi; infinity is their sign), "from about 0.5m to infinity" (raspi.tv) -- and at F a point spreads to 52 px (DERIVED). raspi.tv adds a +2D lens in front, "focus at about 25cm", too far for F; a reader's comment there: "You CAN change the focus of the stock lens on the pi camera. It is tricky but can be done." Unscrew it until the LEDs are sharp.

NOTES (continued)

7. Frame A is the sensor's own 4:3 round A1-A4 and PWR plus 5.00 mm all round. A1 to A4 are in a column on the +Y side of the retention screw, PWR on the -Y side; measured off SQRL's own photograph to +/-1.0 mm. At F the picture is larger; the table gives the crop.
8. OBSERVED, reported by Tim Ansell, 3 Oct 2026, of the deployed hardware: the autofocus camera on the reference Acorn, the "AF-65" OV5647 module, is about 10 cm above the Acorn's LEDs, the picture looks in focus, and it needs a digital zoom or crop onto the Acorn's end. DERIVED, the 10 cm approximate, the module's angles ASSUMED the stock lens's: at 100 mm the picture covers 72.9 x 97.2 mm, frame A is a crop of x 3.97, the 14.47 mm LED column 386 px.
9. S is six terms -- standoffs under the Pi, Pi 5 board, Pi-to-HAT spacers, HAT board, M.2 socket seat, card -- and only the card's 0.80 +/-0.08 is published (M.2 specification); the HAT+ specification only recommends "provide at least 15mm board-to-board spacers; 16mm spacers are ideal". Measure S as one figure.
10. The camera is over the LEDs, not the card: the heatsink's blower stands just short of them, and from here no ray to an LED passes over it; the camera board overhangs it, its underside 5.20 above the lens face.

LENS FACE HEIGHTS, mm: THE LEDs IN VIEW AND IN FOCUS

	THE LENS FACE ABOVE	HEIGHT	TERM
H1	the mounting plate's face, the standoffs' base	$S + 60.0$	S: MEASURE
H2	the assembly's highest point, as clearance	$60.0 - T$	T: MEASURE
F	the card's top face, where the LEDs are	60.0	DERIVED

FRAME A AND THE PICTURE AT THE CARD, mm, DERIVED

LENS TO LEDs	WHAT	COVERS X x Y	CROP TO A	LED COLUMN
--	frame A, long side along Y	18.35 x 24.47	x 1.00	--
F 60.0	the v1.3's picture, refocused	42.6 x 56.9	x 2.32	660 px
about 100	the AF-65's picture, as deployed	72.9 x 97.2	x 3.97	386 px

LEGEND

- Subject outline
- LED, display or connector
- Adjacent part or connector body, and the Acorn CLE-215+
- Frame A's target, on the plane F is measured from
- The 65 deg lens's field of view, 53.50 x 41.41, to its picture's edge
- Frame A, the rectangle the picture must cover
- The mounting plate's face, S below the card: NOT TO SCALE, to measure
- The assembly's highest point, T above the card: NOT TO SCALE, to measure
- Dimension, extension and leader

SOURCES

1. Board geometry: raspberry_pi/boards.py - Pi 5 outline and connectors; see RPI-5.
2. Card, LEDs, HAT and stack: accessories/parts.py - The card, its seat, its LEDs, S and T; see ACC-HAT-M2POE.
3. Raspberry Pi HAT+ specification: <https://web.archive.org/web/20260115212335/https://datasheets.raspberrypi.com/hat/hat-plus-specification.pdf>
4. raspi.tv, 25 May 2013: <https://web.archive.org/web/20260212013652/https://raspi.tv/2013/adapt-your-raspberry-pi-camera-for-close-up-use>
5. Lens figures and focus: RPICAM-LENS - Every lens figure on this sheet, declared and derived, with the vendor pages it is quoted from.

CAMERA OVER THE ACORN CLE-215+

Camera Module v1.3, stock 65 degree lens: lens face heights

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-476-gf39d6ab	mm	1:1, PLAN 1:2
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	RPICAM-OVER-ACORN	A
SUBJECT	GENERAL TOLERANCE	PROJECTION	
Acorn CLE-215+, Pi 5	Pi 5 +/-0.20, LEDs +/-1.0, heights DERIVED		